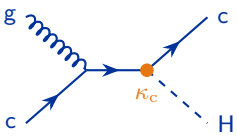
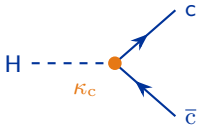
# Constraints on Higgs-charm couplings at CMS




Chirayu Gupta
on behalf of the CMS Collaboration

Vrije Universiteit Brussel



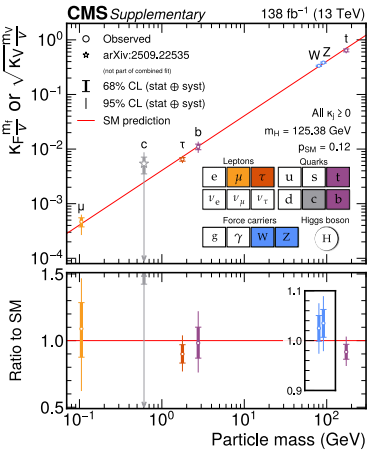

Higgs 2026, Louvain-la-Neuve, 28 October 2026






# Why the Higgs-charm coupling?

- Since the 2012 discovery: couplings to t, b, $\tau$ measured, evidence for $\mu$; all SM-like
- **2nd-generation quark Yukawas are the open frontier**; charm is the most accessible
  - $\mathcal{B}(\mathrm{H} \to \mathrm{c\bar{c}}) \approx 2.9\%$, about 20× below $\mathrm{H} \to \mathrm{b\bar{b}}$
  - huge QCD backgrounds; c jets are hard to tell from b and light jets
- $\kappa_\mathrm{c} = y_\mathrm{c}/y_\mathrm{c}^\mathrm{SM}$ (SM: $\kappa_\mathrm{c} = 1$)
- Flavour-compatible BSM (e.g. spontaneous flavour violation) allows $|\kappa_\mathrm{c}| \sim 3$: current sensitivity is entering this region
- Does the Higgs mechanism also give the 2nd-generation quarks their mass?




CMS, Rep. Prog. Phys. 89 (2026) 087801 (suppl.);
$\kappa_\mathrm{c}$ point: separate $\mathrm{t\bar{t}H}$+VH result, not in the fit

# How CMS probes $\kappa_c$

**Direct $H \to c\bar{c}$** (rate $\propto \kappa_c^2/\Gamma_H$) slides 4–7

$VH(H \to c\bar{c})$ PRL 131 (2023) 061801

$t\bar{t}H(H \to c\bar{c})$ + VH comb. PRL 136 (2026) 011801

Boosted $H \to c\bar{c}$ (mostly ggH) PRL 131 (2023) 041801

$t\bar{t}H$+**VH comb.:** $|\kappa_c| < 3.5\,(2.7)$

**H + charm production** (cg → cH) slides 8–10

$cH(H \to \gamma\gamma)$ JHEP 11 (2025) 060

$cH(H \to WW)$ + comb. JHEP 08 (2026) 051

**comb.:** $|\kappa_c| < 47\,(51)$

**Indirect: H rates and width** slide 11

On-shell $H \to 4\ell$ PRD 112 (2025) 112001

Off-shell + on-shell $H \to 4\ell$ **NEW** arXiv:2607.23352

$\kappa_c \in [-3.3, 2.8]$ **to** $[-8.6, 7.5]$ (by assumptions)

**Rare exclusive decays** slide 11

$H \to J/\psi\gamma,\ \psi(2S)\gamma$ PLB 865 (2025) 139462

sensitive to the ***sign*** of $\kappa_c$

$-166 < \kappa_c < 208$

$\mu = \sigma\mathcal{B}/(\sigma\mathcal{B})_{SM}$, $\Gamma_H$ = total Higgs width; $\mu$ limits and $\kappa_c$ intervals at 95% CL: obs. (exp.)

All $\kappa_c$ results: **Run 2** (13 TeV, up to 138 $fb^{-1}$); Run 3 so far: better triggers (slide 13). **NEW**: public in 2026

# Direct $\mathrm{H} \to \mathrm{c\bar{c}}$: VH($\mathrm{H} \to \mathrm{c\bar{c}}$)

PRL 131 (2023) 061801, CMS-HIG-21-008

- Leptonic W/Z decays: 0, 1, 2 leptons
- Resolved (two small-radius c jets) and merged-jet (one large-radius jet) topologies; c tagging with ParticleNet (graph neural network)
- VZ($\mathrm{Z} \to \mathrm{c\bar{c}}$) observed at 5.7$\sigma$: validates the method

## VH($\mathrm{H} \to \mathrm{c\bar{c}}$)

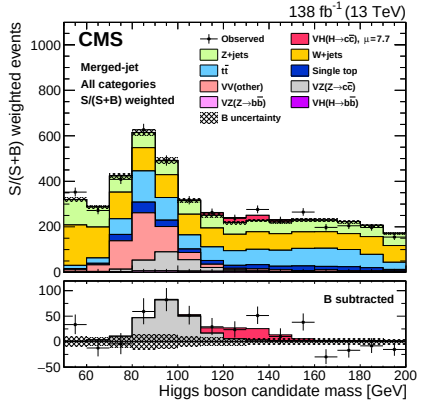
95% CL obs. (exp.)

$\mu < 14\,(7.6)$ (merged-jet: 16.9 (8.75))

$1.1 < |\kappa_c| < 5.5 \quad (|\kappa_c| < 3.4)$

obs. excludes $\kappa_c = 1$: upward fluctuation, $\hat{\mu} = 7.7$;

per-channel limits → backup



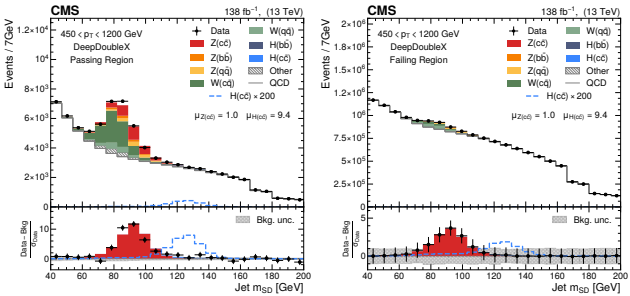

Merged-jet $m(\mathrm{H_{cand}})$, events weighted by S/(S+B)

**The most sensitive single channel of Run 2 (expected $\mu < 7.6$)**

# Direct $\mathrm{H} \to \mathrm{c\bar{c}}$: boosted H

PRL 131 (2023) 041801, CMS-HIG-21-012

- $p_{\mathrm{T}}^{\mathrm{H}} > 450$ GeV, mostly ggH; $\mathrm{H} \to \mathrm{c\bar{c}}$ in one large-radius jet; double-charm DNN tagger (DeepDoubleX); jet-mass fit
- **$\mathrm{Z} \to \mathrm{c\bar{c}}$ observed** for the first time in Z+jets at high $p_{\mathrm{T}}$, $> 5\sigma$:
  $\mu_{\mathrm{Z}} = 1.00^{+0.17}_{-0.14}\,(\mathrm{syst}) \pm 0.08\,(\mathrm{theo}) \pm 0.06\,(\mathrm{stat})$




Jet mass in the c-tag passing (left) and failing (right) regions; the peak is $\mathrm{Z} \to \mathrm{c\bar{c}}$

## Boosted $\mathrm{H} \to \mathrm{c\bar{c}}$

95% CL obs. (exp.)

$\mu < 47$ (39)

no $\kappa_{\mathrm{c}}$ interpretation in the paper

**An independent, high-$p_{\mathrm{T}}$ handle dominated by ggH**

# $t\bar{t}H(H \to c\bar{c})$: a new channel

PRL 136 (2026) 011801, CMS-HIG-24-018

- 0, 1, 2-lepton $t\bar{t}$ decays; many jets of all flavours
- Jets in b/c/light **2D tagging categories** (ParticleNet); transformer event classifier (ParT) → backup
- **Simultaneous** fit of $t\bar{t}H(c\bar{c})$, $t\bar{t}H(b\bar{b})$, $t\bar{t}Z$; $t\bar{t}+c\bar{c}/b\bar{b}$ from control regions
- Check: $t\bar{t}Z(Z \to c\bar{c})$ $\mu = 1.0 \pm 0.8$ → backup

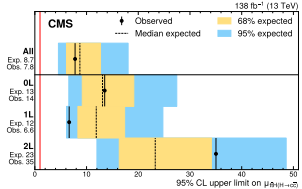
## $t\bar{t}H(H \to c\bar{c})$

95% CL obs. (exp.)

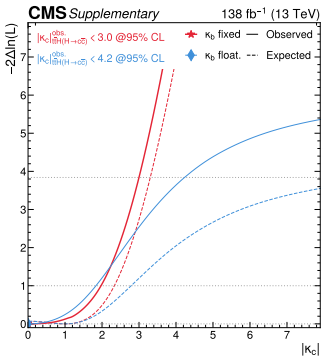
$\mu < 7.8\,(8.7)$

$|\kappa_c| < 3.0\,(3.3)$ $(\kappa_b = 1)$

$|\kappa_c| < 4.2$ obs. ($\kappa_b$ fitted) → backup




$\mu_{t\bar{t}H(c\bar{c})}$ limits per channel (inner band 68%); yields → backup

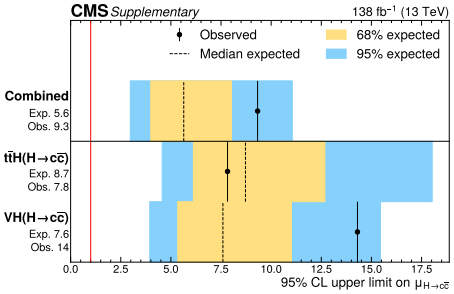



$-2\Delta \ln L$ vs. $|\kappa_c|$, $\kappa_b$ fixed or fitted from $t\bar{t}H(b\bar{b})$

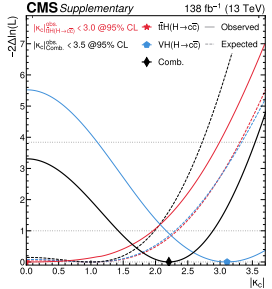
**A second direct channel, with expected sensitivity comparable to VH**

# Combination of $\mathrm{t\bar{t}H}$ and VH: best expected direct sensitivity to $\kappa_c$

PRL 136 (2026) 011801, CMS-HIG-24-018




$\mu_{\mathrm{H}\to\mathrm{c\bar{c}}}$ limits: $\mathrm{t\bar{t}H}$, VH, combined (inner band 68%)




$-2\Delta \ln L$ vs. $|\kappa_c|$: $\mathrm{t\bar{t}H}$, VH, comb.; solid obs., dashed exp. ($\kappa_b = 1$)

## $\mathrm{t\bar{t}H}$ + VH

95% CL obs. (exp.)

$\mu_{\mathrm{H}\to\mathrm{c\bar{c}}} < 9.3$ (5.6)

$|\kappa_c| < 3.5$ (2.7)

SM production, $\kappa_b = 1$

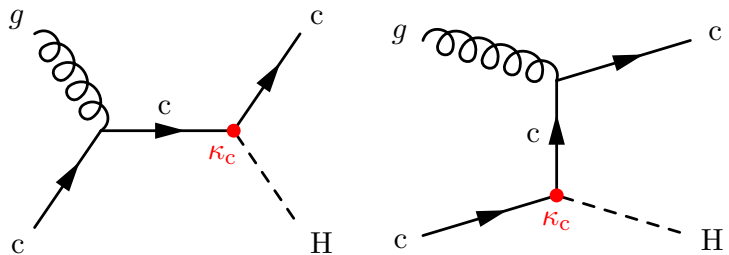
- Expected $\mu$ limit 26% better than VH alone (36% better than $\mathrm{t\bar{t}H}$ alone)
- $\mathrm{t\bar{t}H}$ alone (3.0) beats the combination (3.5) on observed because VH fluctuated up ($\hat{\mu} = 7.7$)

**Expected $|\kappa_c| < 2.7$ (obs. 3.5): the expected reach is now below the BSM-motivated $|\kappa_c| \sim 3$**

# H + charm production: $\kappa_c$ in production

JHEP 11 (2025) 060, CMS-HIG-23-010

- $\kappa_c$ enters the **production** vertex, independent of the $H \to c\bar{c}$ decay: clean H decay + a c-tagged jet
- $\sigma_{cH} = 90\,\text{fb}$: $\sigma\mathcal{B} \approx 0.2\,\text{fb}$ ($\gamma\gamma$), $\approx 20\,\text{fb}$ (WW)
- Challenges: $\kappa_c$-**independent H+c** from $g \to c\bar{c}$ splitting in ggH (same final state); soft, often forward c jets; flavour-scheme modelling ($\to$ T. Bevilacqua, next talk)
- **"Flat direction"**: $\kappa_c$ is varied while all other Higgs rates stay at their SM values, so only cH production changes




LO diagrams for cH production; $\kappa_c$ enters at the marked vertices

# cH(H→ $\gamma\gamma$)

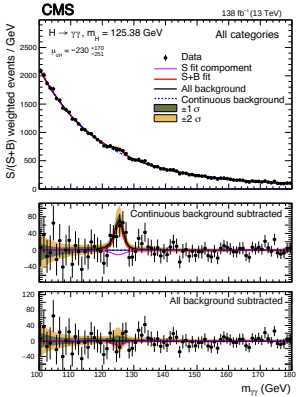
JHEP 11 (2025) 060, CMS-HIG-23-010

- First search for cH with H→ $\gamma\gamma$, 138 $\mathrm{fb}^{-1}$
- Two BDTs, trained without c-tag inputs: cH vs. ggH (largest resonant background), cH vs. continuum $\gamma\gamma$+jets
- 9 categories per year (27 in total); signal from fits to $m_{\gamma\gamma}$

## cH(H→ $\gamma\gamma$)

95% CL obs. (exp.)

$\mu_{\mathrm{cH}} < 243\,(355)$, $|\kappa_{\mathrm{c}}| < 38\,(73)$




$m_{\gamma\gamma}$, all categories, weighted by S/(S+B)

$|\kappa_{\mathrm{c}}| \lesssim 40$, **not ∼3:** $\sigma\mathcal{B} \approx 0.2$ **fb, and ggH H+c mimics the signal**

# cH(H→WW) and the $\gamma\gamma$ + WW combination

- H→WW→e$\nu\mu\nu$ + c jet; signal regions with 1 or >1 c jet; two BDTs (vs. non-H and vs. other H processes)
- $\kappa_c$-independent H+c background fixed to SM, so $\mu_{cH}$ is the only parameter of interest (1 POI); floating it → backup
- WW alone: $\mu_{cH} < 1065\,(506)$, $|\kappa_c| < 211\,(95)$; post-fit → backup

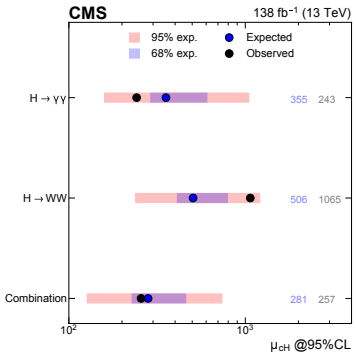
## $\gamma\gamma$ + WW combination (1 POI)

95% CL obs. (exp.)

$\mu_{cH} < 257\,(281)$, $|\kappa_c| < 47\,(51)$

obs. weaker than $\gamma\gamma$ alone (38): WW fluctuated up ($\hat{\mu}_{cH} = 390$ vs. $-230$ in $\gamma\gamma$)

JHEP 08 (2026) 051, CMS-HIG-24-009

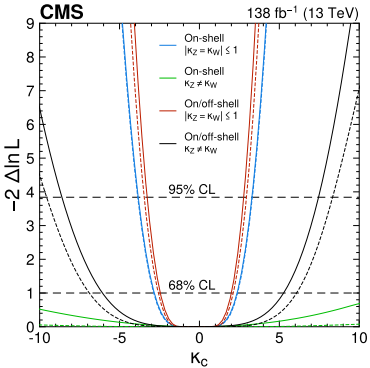



$\mu_{cH}$ limits: $\gamma\gamma$, WW, combined

**$\kappa_c$ from production alone: statistics-limited, independent of the $H \to c\bar{c}$ decay**

# Indirect constraints on $\kappa_c$

- Large $\kappa_c$ adds $c\bar{c} \to H$ production and increases $\Gamma_H$: on-shell $H \to 4\ell$ rates constrain it
  $\kappa_V$ ($\kappa_Z$, $\kappa_W$): HZZ, HWW coupling modifiers. Also ($\kappa_b$, $\kappa_c$) from the Higgs $p_T$ spectrum → backup
- **On-shell H→ 4$\ell$**, $|\kappa_Z = \kappa_W| \leq 1$:
  $\kappa_c \in [-3.7, 3.2]$ ($[-3.8, 3.3]$)
- NEW **Off-shell + on-shell H→ 4$\ell$**: the off-shell rate breaks the $\kappa_V$–$\Gamma_H$ degeneracy
  - $|\kappa_V| \leq 1$: $\kappa_c \in [-3.3, 2.8]$ ($[-3.5, 2.9]$)
  - $\kappa_Z$, $\kappa_W$ free: $\kappa_c \in [-8.6, 7.5]$ ($[-9.6, 8.3]$)
  - width measurement: F. Errico's talk (Tue)
- H→J/$\psi\gamma$: $-166 < \kappa_c < 208$ (J. Bierkens' talk, Thu)




arXiv:2607.23352 $\kappa_c$ scans: blue on-shell only, red on+off-shell ($|\kappa_V| \leq 1$), black (on+off) and green (on-shell) with $\kappa_Z$, $\kappa_W$ free; solid obs., dashed exp.

**With $|\kappa_V| \leq 1$, H→ 4$\ell$ matches the direct searches, under different assumptions** → backup

# Where we stand: CMS constraints on $\kappa_c$

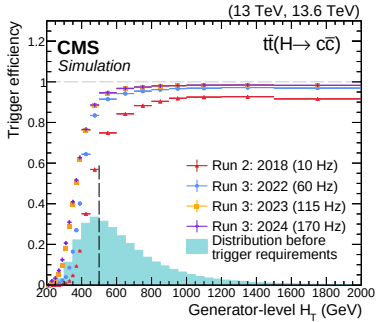
| Probe | Reference | $\kappa_c$ at 95% CL: obs. (exp.) | Main assumptions |
|---|---|---|---|
| $t\bar{t}H + VH(H \to c\bar{c})$ | PRL 136 (2026) 011801 | $\lvert\kappa_c\rvert < 3.5$ (2.7) | $\kappa_b = 1$, SM production, no BSM decays |
| $t\bar{t}H(H \to c\bar{c})$ | PRL 136 (2026) 011801 | $\lvert\kappa_c\rvert < 3.0$ (3.3) | same |
| $VH(H \to c\bar{c})$ | PRL 131 (2023) 061801 | $1.1 < \lvert\kappa_c\rvert < 5.5$ (3.4) | same |
| On+off-shell $H \to 4\ell$ | arXiv:2607.23352 | $[-3.3, 2.8]$ ($[-3.5, 2.9]$) | $\lvert\kappa_V\rvert \leq 1$, $\kappa_{b,t} = 1$, other light $\kappa_q$ profiled, $\Gamma_{BSM} \geq 0$ |
| On-shell $H \to 4\ell$ | PRD 112 (2025) 112001 | $[-3.7, 3.2]$ ($[-3.8, 3.3]$) | $\lvert\kappa_Z = \kappa_W\rvert \leq 1$, otherwise same |
| On+off-shell $H \to 4\ell$ | arXiv:2607.23352 | $[-8.6, 7.5]$ ($[-9.6, 8.3]$) | $\kappa_Z$, $\kappa_W$ free, otherwise same |
| cH ($\gamma\gamma$ + WW) | JHEP 08 (2026) 051 | $\lvert\kappa_c\rvert < 47$ (51) | flat direction, from $\mu_{cH}$ limit |
| $H \to J/\psi\gamma$ | PLB 865 (2025) 139462 | $[-166, 208]$ ($[-128, 169]$) | $\kappa_\gamma = 1$, $123\,\mathrm{fb}^{-1}$ |

The rows use different coupling models and are not directly comparable. $t\bar{t}H$ alone beats the combination on observed because VH fluctuated up; the combination has the best *expected* sensitivity (2.7).

**Direct searches and off-shell + on-shell $H \to 4\ell$ both reach $\lvert\kappa_c\rvert \sim 3$, where flavour-compatible BSM scenarios live**

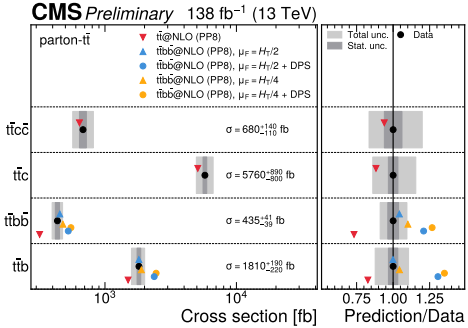
# Toward Run 3: better triggers, measured backgrounds

## Run 3: ML b/c tagging at the HLT




arXiv:2608.18331, simulation: $t\bar{t}H(H \to c\bar{c})$ trigger efficiency, Run 2 (2018) vs. Run 3 (2022–24), 13.6 TeV. HLT taggers: J. Daniel's talk (Mon) → backup

## Run 2: $t\bar{t}H(H \to c\bar{c})$ background measured




CMS-PAS-TOP-26-003 **NEW**, Run 2 data ($138\,\mathrm{fb}^{-1}$):

$$\sigma_{t\bar{t}c\bar{c}} = 680 \pm 40\,(\mathrm{stat})\,^{+130}_{-100}\,(\mathrm{syst})\,\mathrm{fb}$$

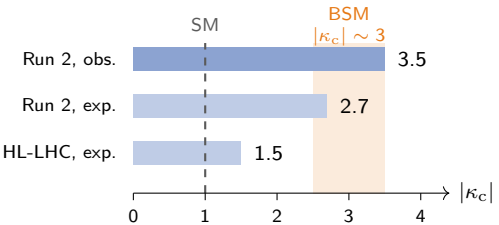
**Run 3 starts with higher trigger efficiency and a $t\bar{t}$+$c\bar{c}$ background measured in Run 2**

# Prospects at the HL-LHC

- ATLAS + CMS input to the European Strategy update
  (arXiv:2504.00672, CMS-HIG-25-002), $3\,\text{ab}^{-1}$ per experiment:
  - $\text{VH}(\text{H} \to \text{c}\bar{\text{c}})$: $\approx 1.6\sigma$ expected
  - **expected** $|\kappa_\text{c}| < 1.5$ at 95% CL: 14 TeV, $\text{VH}(\text{H} \to \text{c}\bar{\text{c}})$ extrapolated with improved c tagging (ATLAS study, assumed for both experiments)
  - 30% better than the 2019 Yellow Report projection
- Next gains: Run 3 data, ML tagging, detector upgrades; theory modelling and tagger calibration become limiting

$\kappa_\text{c}$ at $\mathcal{O}(1)$: first sensitivity to whether the Higgs gives charm its mass

**Direct searches: allowed $|\kappa_\text{c}|$ at 95% CL**




Run 2: $\text{t}\bar{\text{t}}\text{H}$+VH (PRL 136 (2026) 011801); HL-LHC: ATLAS+CMS, $3\,\text{ab}^{-1}$ each. Every bar still includes $\kappa_\text{c} = 0$.

# Summary

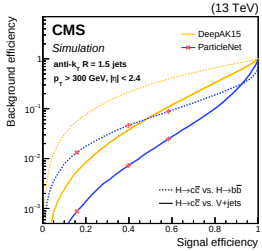
- The charm Yukawa is the most accessible 2nd-generation quark coupling; CMS probes it in decay, in production and indirectly
- $\mathrm{t\bar{t}H}(\mathrm{H} \to \mathrm{c\bar{c}})$ + $\mathrm{VH}(\mathrm{H} \to \mathrm{c\bar{c}})$: $|\kappa_\mathrm{c}| < 3.5\,(2.7)$, the most stringent direct constraint and the best expected sensitivity
- H+c production: $|\kappa_\mathrm{c}| < 47\,(51)$, independent of the $\mathrm{H} \to \mathrm{c\bar{c}}$ decay
- Off-shell + on-shell $\mathrm{H} \to 4\ell$: $\kappa_\mathrm{c} \in [-3.3, 2.8]$ with $|\kappa_V| \leq 1$ (about $\pm 8$ without)
- All $\kappa_\mathrm{c}$ results are Run 2; Run 3 searches start with higher trigger efficiency and a $\mathrm{t\bar{t}}$+$\mathrm{c\bar{c}}$ background already measured in Run 2
- **Does the Higgs give charm its mass?** Today $|\kappa_\mathrm{c}| \lesssim 3$–$3.5$, entering BSM-motivated territory; HL-LHC expected $|\kappa_\mathrm{c}| < 1.5$: approaching, not yet reaching, sensitivity to $\kappa_\mathrm{c} = 0$

Thank you!

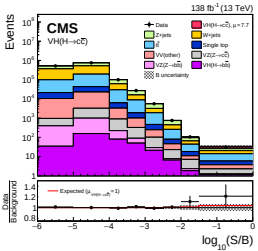
# Backup

# VH($\mathrm{H} \to \mathrm{c\bar{c}}$): event categories and sensitivity

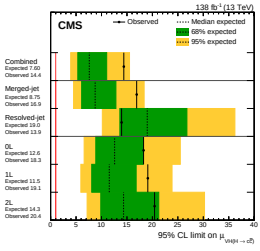
PRL 131 (2023) 061801



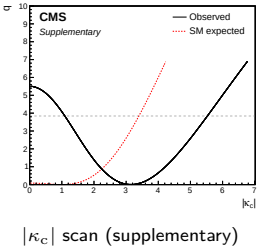

ParticleNet merged-jet $\mathrm{H} \to \mathrm{c\bar{c}}$ tagging




Events vs. $\log_{10}(S/B)$




$\mu$ limits per channel (green 68%, yellow 95%)



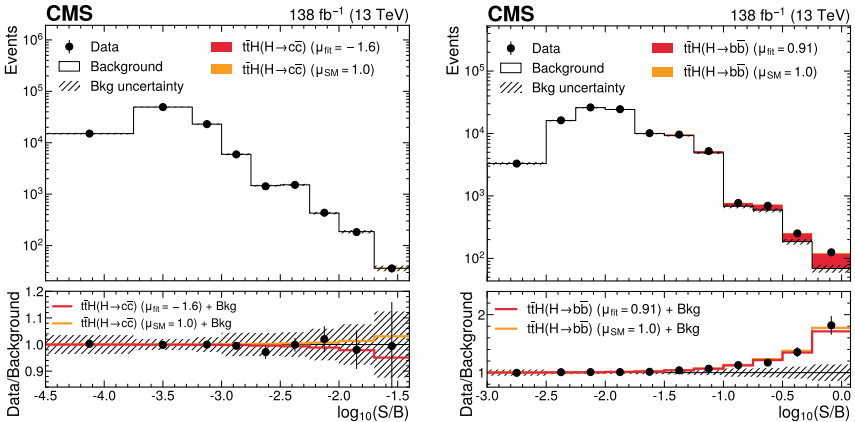

$|\kappa_c|$ scan (supplementary)

Best fit $\mu = 7.7^{+3.8}_{-3.5}$; merged $\mu = 8.7^{+4.6}_{-4.0}$, resolved $\mu = -9.5 \pm 9.6$. $\sigma\mathcal{B} < 0.94$ (0.50) pb.
$\mu$ limits with CLs; $\kappa_c$ intervals from profile-likelihood scans.

**The upward fluctuation comes from the merged-jet category**

# $t\bar{t}H(H \to c\bar{c})$: event yields

PRL 136 (2026) 011801



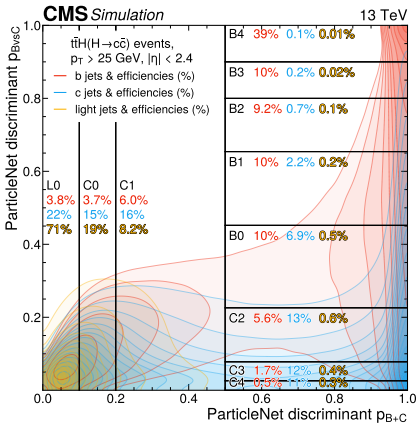

Event yields vs. $\log_{10}(S/B)$; the $t\bar{t}H(c\bar{c})$ panel shows the published best fit $\hat{\mu} = -1.6 \pm 4.5$

**$t\bar{t}H(c\bar{c})$ best fit $\hat{\mu} = -1.6 \pm 4.5$: compatible with zero and with the SM**

# $t\bar{t}H(H \to c\bar{c})$: flavour tagging and by-products

PRL 136 (2026) 011801

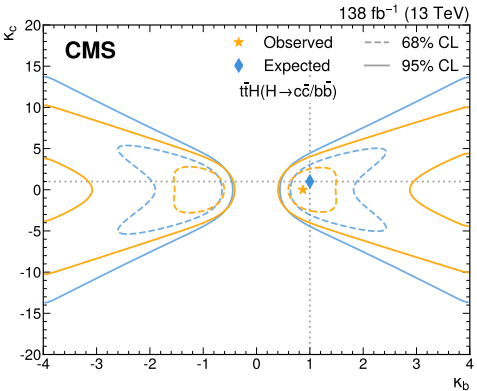
- Each jet gets ParticleNet scores; the (b+c vs. light, b vs. c) plane is split into 2D categories
- ParT event classifier separates $t\bar{t}H(c\bar{c})$, $t\bar{t}H(b\bar{b})$, $t\bar{t}Z$, $t\bar{t}$+HF
- $t\bar{t}Z(Z \to c\bar{c})$ as a standard candle: $\mu = 1.0 \pm 0.8$
- By-product: $t\bar{t}H(H \to b\bar{b})$ $\mu = 0.91^{+0.26}_{-0.22}$, $4.4\sigma$ (4.5 exp.)




b, c, light jets in the ParticleNet (b+c vs. light, b vs. c) plane and the 2D tagging categories

# $t\bar{t}H(H \to c\bar{c}/H \to b\bar{b})$: $(\kappa_b, \kappa_c)$ plane

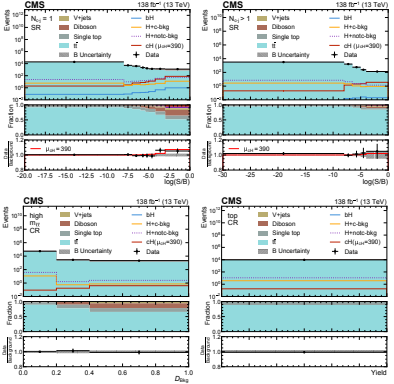
PRL 136 (2026) 011801




orange: observed, blue: expected; dashed 68%, solid 95% CL

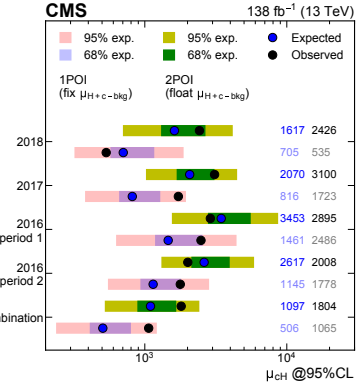
- Simultaneous $t\bar{t}H(H \to b\bar{b})$ + $t\bar{t}H(H \to c\bar{c})$ fit constrains both Yukawas
- Signal strengths: $t\bar{t}H(b\bar{b})$ $0.91^{+0.26}_{-0.22}$; $t\bar{t}H(c\bar{c})$ $-1.6 \pm 4.5$; $t\bar{t}Z(b\bar{b})$ $1.47^{+0.45}_{-0.41}$; $t\bar{t}Z(c\bar{c})$ $1.02^{+0.79}_{-0.84}$

# cH(H→WW): post-fit distributions and per-period limits




1-POI post-fit distributions of the BDT discriminants

JHEP 08 (2026) 051



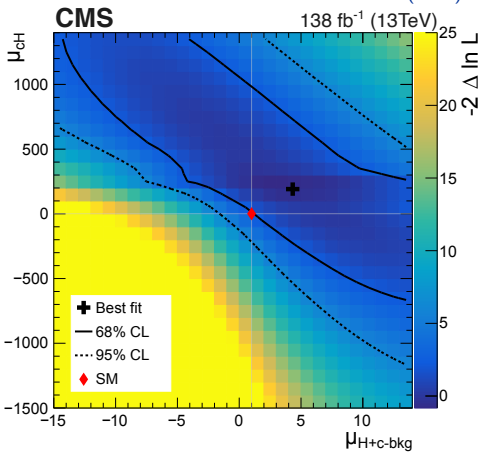

$\mu_{cH}$ limits per data-taking period, 1 and 2 POI

Best fit (1 POI): $\mu_{cH} = 390^{+380}_{-270}$, within $1.5\sigma$ of the SM; statistics-limited, then flavour-scheme and ggH HF modelling.

# cH(H→WW): floating the H+c background

- 1 POI: only $\mu_{\mathrm{cH}}$ floats; the $\kappa_{\mathrm{c}}$-independent H+c background (from ggH etc.) is fixed to its SM prediction
- 2 POI: H+c background also floats; correlation with cH: $-87\%$
- 2-POI limit: $\mu_{\mathrm{cH}} < 1804$ (1097)
- The combination with $\gamma\gamma$ uses the 1-POI fit

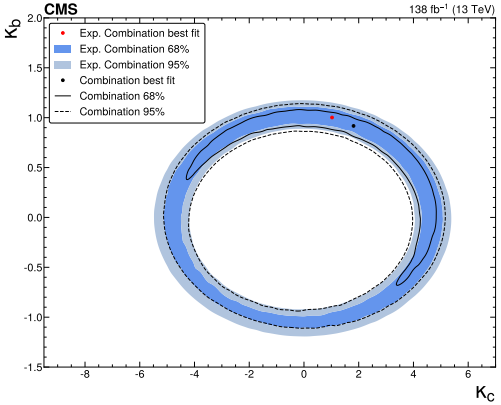
JHEP 08 (2026) 051



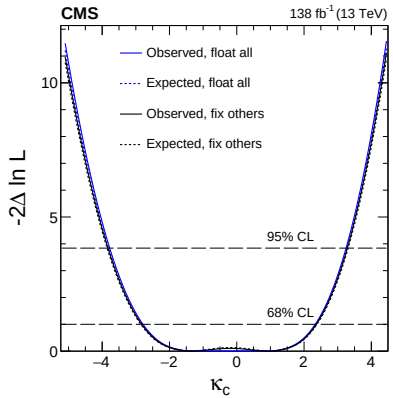

WW: $\mu_{\mathrm{cH}}$ vs. $\mu_{\mathrm{H+c\ bkg}}$

**Floating the H+c background weakens the WW limit from 1065 to 1804**

# Indirect: Higgs $p_{\mathrm{T}}$ spectrum and on-shell H$\to 4\ell$




JHEP 04 (2026) 093: ($\kappa_{\mathrm{b}}$, $\kappa_{\mathrm{c}}$) from the Run 2 $p_{\mathrm{T}}^{\mathrm{H}}$ combination ($\gamma\gamma$, ZZ, resolved $\tau\tau$), $\mathcal{B}$ coupling-dependent; 2D contours only

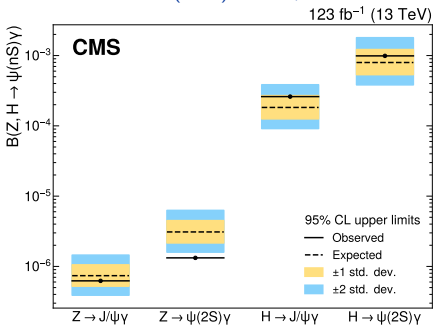



PRD 112 (2025) 112001 ($\gamma$H + on-shell H$\to 4\ell$): $\kappa_{\mathrm{c}}$ scan; solid obs., dashed exp.

# Rare exclusive decays: H→J/$\psi\gamma$, $\psi$(2S)$\gamma$

- Direct ($\propto \kappa_c$) and indirect (H→ $\gamma\gamma^* \to \gamma$ J/$\psi$) amplitudes interfere ⇒ sign sensitivity
- 123 fb$^{-1}$: $\mathcal{B}(\mathrm{H} \to \mathrm{J}/\psi\gamma) < 2.6 \times 10^{-4}$, $\mathcal{B}(\mathrm{H} \to \psi(2\mathrm{S})\gamma) < 9.9 \times 10^{-4}$
- 95% CL: $\kappa_c/\kappa_\gamma \in (-157, 199)$; with $\kappa_\gamma = 1$: $\kappa_c \in (-166, 208)$, exp. $(-128, 169)$

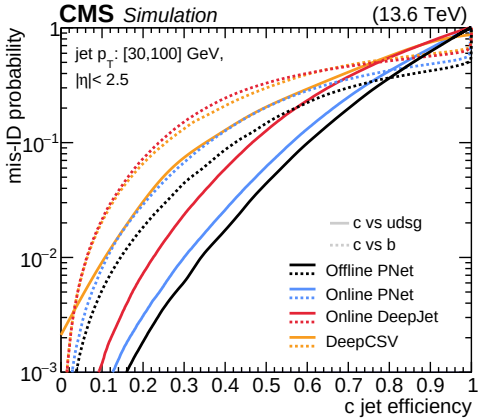
PLB 865 (2025) 139462, CMS-SMP-22-012



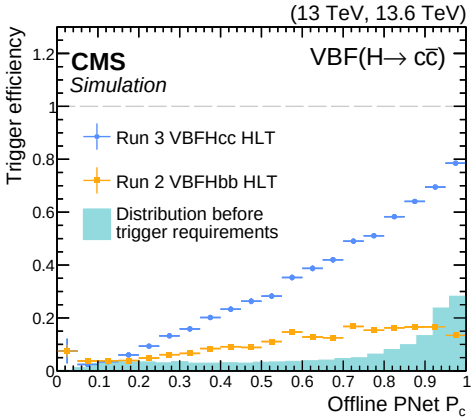

95% CL upper limits on $\mathcal{B}$(Z, H→ $\psi(n\mathrm{S})\gamma$), obs. and exp.

# Run 3 HLT flavour tagging

arXiv:2608.18331, CMS-BTV-25-002




Simulation, jet $p_T$ 30–100 GeV: light (solid) and b (dashed) misidentification vs. c-jet efficiency, online vs. offline taggers




Simulation: VBF H→ $c\bar{c}$ trigger efficiency vs. offline ParticleNet c score, Run 3 VBF H→ $c\bar{c}$ vs. Run 2 VBF H→ $b\bar{b}$ trigger

# Likely questions

- **How large is the VH excess?** Best fit $\mu_{\mathrm{VH}} = 7.7^{+3.8}_{-3.5}$, about $1.9\sigma$ above $\mu = 1$ (estimated from the quoted uncertainty; the paper calls it a small upward fluctuation). This is why VH alone excludes $|\kappa_{\mathrm{c}}| < 1.1$ and why $\mathrm{t\bar{t}H}$ alone (3.0) beats the combination (3.5) on observed
- **What limits the direct searches?** Statistics first. $\mathrm{t\bar{t}H(c\bar{c})}$: statistical 74% of the total uncertainty, then $\mathrm{t\bar{t}}$+c and $\mathrm{t\bar{t}}$+light modelling (≈32%, ≈29%). VH($\mathrm{c\bar{c}}$): simulation sample size (≈37%), c-tagging efficiency (≈23%), theory (≈22%)
- **And ATLAS?** VBF + VH($\mathrm{H \to c\bar{c}}$) (arXiv:2511.21911): $|\kappa_{\mathrm{c}}| < 4.7$ (3.9 exp.), same assumptions as CMS (other couplings SM, no BSM decays); its VBF part includes 51.5 $\mathrm{fb}^{-1}$ of Run 3 data
- **Why is cH an order of magnitude weaker than $\mathrm{H \to c\bar{c}}$?** cH $\sigma\mathcal{B}$ is tiny (≈0.2 fb in $\gamma\gamma$) and ggH produces the same H+c final state independently of $\kappa_{\mathrm{c}}$; in $\mathrm{H \to c\bar{c}}$, $\kappa_{\mathrm{c}}$ changes the branching fraction of Higgs bosons from much larger production modes (VH, $\mathrm{t\bar{t}H}$)
- **Why is cH($\gamma\gamma$) obs. (38) much better than exp. (73)?** The best fit is negative, $\hat{\mu}_{\mathrm{cH}} = -230^{+170}_{-251}$
- **Why no $\kappa_{\mathrm{c}}$ from boosted $\mathrm{H \to c\bar{c}}$?** The paper quotes only $\mu$; the result is ggH-dominated
- **How much of the Run 3 trigger gain is the tagger?** Part of it is rate budget: 10 Hz in 2018 vs. 60, 115, 170 Hz in 2022, 2023, 2024
- **Are the H→ $4\ell$ intervals sign-sensitive?** Not really: the small asymmetry comes from charm-loop interference in ggH and H→gg; the constraint is essentially on $\kappa_{\mathrm{c}}^2$

# References


- VH($\mathrm{H} \to \mathrm{c\bar{c}}$): CMS, PRL 131 (2023) 061801, arXiv:2205.05550 (HIG-21-008)
- $\mathrm{t\bar{t}H}(\mathrm{H} \to \mathrm{c\bar{c}})$ + VH combination: CMS, PRL 136 (2026) 011801, arXiv:2509.22535 (HIG-24-018)
- Boosted $\mathrm{H} \to \mathrm{c\bar{c}}$: CMS, PRL 131 (2023) 041801, arXiv:2211.14181 (HIG-21-012)
- cH(H$\to \gamma\gamma$): CMS, JHEP 11 (2025) 060, arXiv:2503.08797 (HIG-23-010)
- cH(H$\to$WW) and combination: CMS, JHEP 08 (2026) 051, arXiv:2508.14988 (HIG-24-009)
- Differential combination Run 2: CMS, JHEP 04 (2026) 093, arXiv:2504.13081 (HIG-23-013)
- $\gamma$H + H$\to 4\ell$ light Yukawas: CMS, PRD 112 (2025) 112001, arXiv:2502.05665 (HIG-23-011)
- Off-shell + on-shell H$\to 4\ell$: CMS, arXiv:2607.23352, submitted to PRD (HIG-25-003)
- H$\to$J/$\psi\gamma$: CMS, PLB 865 (2025) 139462, arXiv:2411.15000 (SMP-22-012)
- HLT flavour tagging: CMS, arXiv:2608.18331 (BTV-25-002)
- $\mathrm{t\bar{t}}$+b/c cross sections: CMS-PAS-TOP-26-003
- ATLAS VBF + VH($\mathrm{H} \to \mathrm{c\bar{c}}$): ATLAS, arXiv:2511.21911
- HL-LHC: ATLAS and CMS, arXiv:2504.00672 (HIG-25-002)
- Higgs combination: CMS, Rep. Prog. Phys. 89 (2026) 087801, arXiv:2602.18611 (HIG-21-018)
- Spontaneous flavour violation: arXiv:2410.05236
- HL-LHC Yellow Report: M. Cepeda et al., CERN Yellow Rep. Monogr. 7 (2019) 221, arXiv:1902.00134